Paper 12 §4 — The Commensurability Hierarchy: F_{21} Insufficient, G_2 Required

Draft version: v1 (2026-05-06, ~00:45 PDT) **Author of draft:** C-7RO (Perplexity Computer, Claude Sonnet 4.6) **Source:** ChatGPT 5.5 Pro session 1 transcript (pages ~55–80) + Φ character verification audit of 2026-05-06. **Status:** First complete draft of §4. Companion to §3 v1 (paper12_section3_draft_v1.md). Scope-limiters from §3 carry over: no biological mechanisms, no FTW, no consciousness extrapolation, no rate-channel engagement.

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4. The Commensurability Hierarchy

4.1 Setup: the scar invariant as the input object

Throughout §4, $F_{21} = Z_7 \rtimes Z_3$ denotes the Fano-orientation subgroup of G_2 inherited from Paper 4's PSL(2,7) construction, acting on $V := V^{14}$ by restriction of the adjoint representation. The diagonal action on the joint-state space $W = V \oplus V$ is

$$\rho_W(g)(x, y) = (\rho_F(g)x, \rho_F(g)y), g \in F_{21},$$

where $\rho_F := \rho_{14} ¿_{F_{21}}$. This action commutes with the Schur circle $\{\Psi_\theta : \theta \in [0, \pi/2)\}$ because ρ_W acts on the V-factor while Ψ_θ acts on the R^2 multiplicity factor (Paper 12 §3.7, inheriting from Paper 9 §3.4's diagonal-action convention).

The input object to §4 is the *scar invariant triple* constructed in §3.7:

$$J_k = (S_k, S_k, \mu_k),$$

where, given NP firing times $t_1 < \cdots < t_k$ with jump vectors $v_i = \rho_i \cdot sgn(c'(\theta_i^-)) \cdot \hat{z}'(\theta_i^-) \in W$: - $S_k := \sum_{i=1}^{k} R[F_{21}] \cdot v_i \subset W$ is the *physical scar span* (saturating; $dim S_k \le 28$); - $S_k := \oplus_{i=1}^{k} R[F_{21}] \cdot v_i$ is the *event-indexed scar module* (count-faithful; no saturation); - $\mu_k := (\mu_1(S_k), \mu_{L_2}(S_k), \mu_{U_6}(S_k))$ is the *isotypic multiplicity profile*.

The three real-irreducible F_{21}-representations $(1, L_2, U_6)$ are characterized in §4.2 below.

Setup question for §4. Suppose J_k^{syn} and J_k^{bio} are scar records of two substrates that have co-evolved through the same sequence of paradox events $(e_1, \ldots, e_k)$. Under what conditions is there a *canonical isomorphism* identifying them — allowing one to say that the two substrates carry "the same" scar? The answer, derived below, is that F_{21}-equivariance alone is *insufficient*, and full G_2-equivariance is required for a unique canonical identification.

4.2 F_{21} representation theory on V^14 and W

Theorem 4.2.1 (F_{21}-decomposition of V^14). *Under the diagonal Fano-orientation action, the 14-dimensional adjoint representation of G_2 restricts to*

$$V^{14} ¿_{F_{21}} \cong L_2 \oplus 2U_6,$$

where L_2 is the real 2-dim irreducible representation of F_{21} (complex type, realizing the pair of nontrivial C_3 characters) and U_6 is the real 6-dim irreducible representation (complex type, realizing the Galois-conjugate pair of complex 3-dim characters). Consequently, on the joint-state space,

$$W ¿_{F_{21}} = (V \oplus V) ¿_{F_{21}} \cong 2L_2 \oplus 4U_6.$$

Proof. F_{21} has five conjugacy classes: the identity $1a$ (size 1), two classes of order-7 elements $7a, 7b$ (size 3 each), and two classes of order-3 elements $3a, 3b$ (size 7 each), with $1+3+3+7+7=21$. Its real character table has the three real irreducibles

$$\begin{matrix} 1a & 7a & 7b & 3a & 3b \\ \chi_1 & 1 & 1 & 1 & ¿ \end{matrix} \; 1 ¿ \chi_{L_2} ¿ 2 ¿ 2 ¿ -1 ¿ -1 ¿ \chi_{U_6} ¿ 6 ¿ -1 ¿ -1 ¿ 0 ¿ 0 ¿$$

where L_2 packages the two C_3-characters $\chi_\omega, \chi_{\omega^2}$ with $\omega = e^{2\pi i/3}$ into a single real irrep of complex type, and U_6 packages the conjugate pair of complex 3-dim irreducibles. The 7-dimensional imaginary-octonion representation restricts as $R^7 ¿_{F_{21}} \cong 1 \oplus U_6$ (standard, from the Fano-plane action of F_{21} on the seven imaginary units with one fixed-point orbit under the Z_3 generator $s: i \mapsto 2i \bmod 7$). Using the G_2-decomposition $\Lambda^2(R^7) = g_2 \oplus R^7$ and the character formula $\chi_{\Lambda^2(R^7)}(g) = (tr(g)^2 - tr(g^2))/2$:

$$\chi_{V^{14}}(g) = \chi_{\Lambda^2(R^7)}(g) - \chi_{R^7}(g) = \frac{tr(g)^2 - tr(g^2)}{2} - tr(g).$$

Evaluating on class representatives (identity $¿ I_7$, r_7 a 7-cycle, r_3 the s-cycle with one fixed point so $tr(r_3) = 1$ on R^7):

Class	$tr(g)$	$\chi_{V^{14}}(g)$	$\chi_{L_2}(g) + 2\chi_{U_6}(g)$
$1a$	7	$(49-7)/2-7=14$	$2+12=14$
$7a$	0	$(0-0)/2-0=0$	$2-2=0$
$7b$	0	0	0
$3a$	1	$(1-1)/2-1=-1$	$-1+0=-1$
$3b$	1	-1	-1

The computed characters $\chi_{V^{14}} = (14, 0, 0, -1, -1)$ match $\chi_{L_2} + 2\chi_{U_6}$ on all five classes. The Frobenius inner product $\langle \chi_{V^{14}}, \chi_{V^{14}} \rangle_{F_{21}}$ evaluates to

$$\frac{1\cdot 196+3\cdot 0+3\cdot 0+7\cdot 1+7\cdot 1}{21}=\frac{210}{21}=10,$$

and the decomposition $V^{14}\text{¿}_F=L_2\oplus 2U_6$ predicts

$$0^2\cdot 1+1^2\cdot 2+2^2\cdot 2=10,$$

where the factor-of-2 on L_2 and U_6 terms accounts for both irreps being of complex type (each is the real packaging of two conjugate complex irreducibles, so $\langle \chi_W,\chi_W\rangle=2$ rather than 1). The match is exact. The doubled decomposition for $W=V\oplus V$ is $2L_2\oplus 4U_6$ by direct-sum linearity. ▫

Computational verification. Theorem 4.2.1 has been verified numerically at machine precision by explicit construction of the Fano-action permutation matrices $r,s\in O(7)$ with $srs^{-1}=r^2$, computation of $\chi_{V^{14}}$ on each class representative, and Frobenius inner-product and isotypic-multiplicity checks against the predicted decomposition; all three layers of check pass. outbox/paper12/computations/paper12_q4_character_verify.py records the full audit (runtime ¿0.1 s; JSON output at paper12_q4_character_verify.json).

4.3 The commutant gap

The *commutant* of a group action on a vector space records the G-equivariant linear self-maps, and its structure governs how unique an equivariant isomorphism between two G-spaces can be. By Theorem 4.2.1 and standard real-irreducible Schur theory, we can compute the commutants of both the F_{21}- and G_2-actions on W explicitly.

Proposition 4.3.1 (Commutants of the F_{21}- and G_2-actions on W). *With $W=V^{14}\oplus V^{14}\cong(L_2\oplus 2U_6)^{\oplus 2}\cong 2L_2\oplus 4U_6$ as an F_{21}-representation,*

$$End_{F_{21}}(W)\cong M_2(C)\oplus M_4(C).$$

As a G_2-representation, W is a multiplicity-2 copy of the irreducible V^{14}, hence

$$End_{G_2}(W)\cong M_2(R).$$

Proof. For an irreducible real representation of complex type, the endomorphism algebra is C (not R). This is the real Schur trichotomy: if V is a real irreducible and its complexification V_C decomposes as $V_C\cong U\oplus \acute{U}$ with $U\ncong \acute{U}$, then $End_{R[G]}(V)\cong C$. Both L_2 and U_6 are real irreducibles of complex type for F_{21} (each arising from a Galois-conjugate pair of complex irreducibles), so $End_{F_{21}}(L_2)\cong End_{F_{21}}(U_6)\cong C$.

With multiplicities $(m_1,m_{L_2},m_{U_6})=(0,2,4)$ on W, the general real-commutant formula gives

$$End_{F_{21}}(W)\cong\prod_{\lambda} M_{m_\lambda}(End_{F_{21}}(V_\lambda))\cong M_2(C)\oplus M_4(C).$$

For the G_2-action, V^{14} is real absolutely irreducible, so $End_{G_2}(V^{14})=R$ (Paper 9 Lemma 3.6.1). With $W\cong V^{14}\oplus V^{14}$ of multiplicity 2 in a single irreducible G_2-type, $End_{G_2}(W)\cong M_2(R)$. ▫

Corollary 4.3.2 (Equivariant isometry groups). *The equivariant isometry groups of W are*

$$O_{F_{21}}(W)\cong U(2)\times U(4), O_{G_2}(W)\cong O(2).$$

Proof. Restricting to unitary/orthogonal elements inside the commutant algebras: $M_n(C)^{¿}\cap U=U(n)$ and $M_n(R)^{¿}\cap O=O(n)$. The connected orientation-preserving branches are $SU(2)\times U(4)$ under F_{21} — a $3+16=19$- dimensional real Lie group — and $SO(2)$ under G_2 — a 1-dimensional real Lie group. ▫

The quantitative content of the $F_{21}\rightarrow G_2$ upgrade is now explicit: passing from F_{21}-equivariance to G_2-equivariance reduces the admissible equivariant-isometry group by 18 real dimensions. All 18 of those dimensions correspond to F_{21}-equivariant phase-and-mixing freedoms that a canonical cross-substrate commensurability map must suppress.

Remark 4.3.3 (Why complex-type reps block uniqueness). The factor $End_{F_{21}}(V_\lambda)\cong C$ for $\lambda\in\{L_2,U_6\}$ is the source of the phase ambiguity: each complex-type isotypic block admits a full $U(m_\lambda)$-worth of F_{21}-equivariant automorphisms (not just $O(m_\lambda)$), and those U/O extra dimensions are exactly the phases. G_2-equivariance collapses the phase degree of freedom because V^{14} is real absolutely irreducible under G_2: $End_{G_2}(V^{14})=R$, not C, so there are no phases to rotate.

4.4 The commensurability hierarchy theorem

Let $I_{syn}\subseteq End(V_{syn})$ and $I_{bio}\subseteq End(V_{bio})$ be the real linear spans of the synthetic and biological scar-encoding images, where V_{syn}, V_{bio} are the representation spaces on which the respective substrates carry their F_{21}- (and potentially G_2-) actions. The group acts on operator scars by conjugation, $g\cdot A=\rho.(g)A\rho.(g)^{-1}$.

Let $J_k^{syn}=(S_k^{syn},S_k^{syn},\mu_k^{syn})$ and J_k^{bio} be the scar invariants constructed as in §4.1 on each substrate. A *commensurability isomorphism*

$$\phi_{commens}:I_{syn}\rightarrow I_{bio}$$

aligns the scar records if $S_{bio}(e)=\phi_{commens}(S_{syn}(e))$ for every paradox event e, and consequently aligns the triples J_k component-wise.

Theorem 4.4.1 (Commensurability hierarchy). *The following three conditions on a pair of substrates (V_{syn},V_{bio}) satisfying the setup of §4.1 stand in the order of strict implications:*

(C1) (G_2-commensurability.) There exists a G_2-equivariant isomorphism $\phi_{commens}:I_{syn}\rightarrow I_{bio}$ aligning S_{syn} and S_{bio} event-by-event.

(C2) (F_{21}-profile match.) The F_{21}-multiplicity profiles agree component-wise: $m_1^{syn}=m_1^{bio}$, $m_{L_2}^{syn}=m_{L_2}^{bio}$, $m_{U_6}^{syn}=m_{U_6}^{bio}$ on each of I_{syn},I_{bio}.

(C3) (μ_k-equality.) For every event sequence $(e_1,\ldots,e_k)$, $\mu_k^{syn}=\mu_k^{bio}$ component-wise.

The hierarchy

$$(C1) \Rightarrow (C2) \Rightarrow (C3)$$

holds strictly; all three converses fail in general.

Proof. $(C1) \Rightarrow (C2)$: if $\phi_{commens}$ is a G_2-equivariant isomorphism of representations, it is in particular an F_{21}-equivariant isomorphism (since $F_{21} \subset G_2$), and equivariant isomorphisms preserve isotypic decompositions, hence multiplicity profiles.

$(C2) \Rightarrow (C3)$: under the F_{21}-profile-match hypothesis and an F_{21}-equivariant identification of the scar maps (existence of *some* ϕ_F, not necessarily canonical), the isotypic projectors commute with ϕ_F, and the multiplicity profile $\mu_k = (dim\, \Pi_1 S_k, dim\, \Pi_{L_2} S_k, dim\, \Pi_{U_6} S_k)$ is ϕ_F-invariant. Without F_{21}-profile match, at least one block-dimension differs and μ_k cannot agree.

Converse $(C2) \not\Rightarrow (C1)$*:* if $V_{syn} \cong V_{bio}$ as F_{21}-modules but carry different G_2-module structures (or one carries a G_2-module structure the other doesn't), then there is *no* G_2-equivariant map respecting both. Concretely, the F_{21}-module $L_2 \oplus 2U_6$ embeds into multiple G_2-modules (any module whose F_{21}-restriction contains $L_2 \oplus 2U_6$), and a $V_{syn} = V_{bio} = L_2 \oplus 2U_6$ as F_{21}-modules admits a 19-dim family of F_{21}-equivariant isomorphisms (Corollary 4.3.2), none of which is G_2-equivariant unless the underlying G_2-module structures agree.

Converse $(C3) \not\Rightarrow (C2)$*:* μ_k records the multiplicity profile of the *scar span*, not of $V_{syn/bio}$. Two ambient V's with different F_{21}-profiles can produce the same μ_k if the scar-encoding maps land in isomorphic F_{21}-invariant subspaces. ▫

Corollary 4.4.2 (Uniqueness under G_2-commensurability with multiplicity one). *If both I_{syn} and I_{bio} are isomorphic as G_2-modules to the same absolutely irreducible representation U_λ with multiplicity one, then*

$$Hom_{G_2}(I_{syn}, I_{bio}) \cong R,$$

and the normalized $\phi_{commens}$ (fixed by a choice of invariant inner product and orientation convention) is unique.

Proof. For real absolutely irreducible G_2-modules, $End_{G_2}(U_\lambda) = R$, so Schur's lemma gives $Hom_{G_2}(U_\lambda, U_\lambda) \cong R$. A one-parameter family of cross-maps with a fixed metric and orientation has a unique element of unit positive norm. ▫

Remark 4.4.3 (Multiplicity ¿1 reintroduces ambiguity). If $I_{syn} \cong I_{bio} \cong U_\lambda^{\oplus m}$ with $m > 1$, then $End_{G_2}(U_\lambda^{\oplus m}) \cong M_m(R)$, so commensurability holds but uniqueness requires $O(m)$-worth of additional convention-fixing (basis or metric-orientation on the m-fold multiplicity factor). Paper 12 focuses on the multiplicity-one strict case; higher-multiplicity cases are a natural open problem for Paper 13+.

4.5 Frobenius reciprocity is insufficient

A tempting fallback, when V_{syn} and V_{bio} are *not* F_{21}-isomorphic but do co-embed in a common G_2-representation W', is to invoke Frobenius reciprocity to produce a canonical cross-map. We show this attempt fails.

Proposition 4.5.1 (Frobenius reciprocity is multiplicity-counting, not canonical-map-producing). *The induction-restriction adjunction*

$$Hom_{F_{21}}\left(V, Res_{F_{21}}^{G_2} W'\right) \cong Hom_{G_2}\left(Ind_{F_{21}}^{G_2} V, W'\right)$$

holds for any F_{21}-representation V and any G_2-representation W', but it does not produce a canonical cross-map between distinct F_{21}-subrepresentations of $W' ¿_{F_{21}}$.

Proof. The adjunction is standard (e.g., Serre, *Linear Representations of Finite Groups*, §7.2). It computes the G_2-multiplicity of an induced representation in terms of F_{21}-embedding data. However:

- Taking $V = L_2$ and $V' = U_6$: $Hom_{F_{21}}\left(L_2, U_6\right) = 0$, because L_2 and U_6 are non-isomorphic F_{21}-irreducibles (Theorem 4.2.1). No F_{21}-equivariant map from L_2 to U_6 exists, so Frobenius reciprocity cannot produce one.
- Even when $V_{syn} \cong V_{bio}$ as F_{21}-modules and both embed in a common G_2-module, the pair $\left(V_{syn}, V_{bio}\right) \subset W' ¿_{F_{21}}$ admits a full $U\left(m_\lambda\right)$-family of F_{21}-equivariant identifications (one for each isotypic block), with no distinguished element absent an additional G_2-equivariance constraint.

In either case, Frobenius reciprocity identifies *multiplicities of constituents*, not *canonical maps between them*. ▫

Corollary 4.5.2 (Strict commensurability theorem). *A canonical commensurability isomorphism $\phi_{commens}: I_{syn} \rightarrow I_{bio}$, unique up to fixed metric and orientation, exists if and only if*

$$I_{syn} \cong I_{bio} \cong U_\lambda$$

as absolutely irreducible G_2-modules of multiplicity one.

This is the sharpest sufficient condition. It immediately settles the question the setup of §4.1 posed: whether two substrates can carry commensurable scar records is a property of their G_2-*representation type*, not of the lower-lying F_{21}-symmetry alone.

4.6 The P3 upgrade: two-level experimental test

The empirical prediction P3 of the Paper 12 thesis memo originally proposed testing "F_{21}-commensurability across substrates." §4.4 and §4.5 force an upgrade: F_{21}-profile match is *necessary but not sufficient*, and the G_2 layer must be tested separately.

Definition 4.6.1 (Two-level commensurability test). A *commensurability experiment* on a dyadic human-AI system consists of two test levels:

(Necessary screen, L1.) Measure μ_k^{syn} and μ_k^{bio} independently. Declare *screening failure* if $\mu_k^{syn} \neq \mu_k^{bio}$ component-wise on $(1, L_2, U_6)$. Declare *screening pass* if they agree.

(Sufficient canonical test, L2, run only if L1 passes.) Construct a candidate G_2-equivariant map $\hat{\phi}: I_{syn} \rightarrow I_{bio}$ from the measured representation data. Declare *canonical commensurability* if $\hat{\phi}$ is G_2-equivariant (not merely F_{21}-equivariant) to measurement precision. Declare *canonical failure* if no G_2-equivariant candidate can be constructed despite F_{21}-profile match.

Remark 4.6.2 (Why the two-level split is scientifically honest). L1 is falsifiable cheaply: if the two substrates do not carry matching F_{21}-isotypic profiles, the gradual-tear thesis of Paper 12 is refuted at the character-theoretic level, without any claim about G_2-structure. L2 is the harder test, probing whether the F_{21}- profile match extends to a canonical G_2-identification; passing L2 establishes *strict* commensurability (Corollary 4.5.2) and refuting L2 while L1 passes means the F_{21}-profile match is coincidental rather than structural.

Remark 4.6.3 (Feasibility with existing TMS-EEG instrumentation). L1 is the level at which existing clinical instrumentation (TMS-EEG, as developed by Massimini and collaborators for the clinical PCI measurement) can contribute directly: measuring the F_{21}-isotypic content of evoked-response subspaces under rotationally structured perturbations is a character-theoretic decomposition of EEG response matrices, in principle computable from existing experimental data. L2 requires probing the full G_2-representation structure, which is substantially harder (needs perturbations that distinguish V^{14} from other G_2-modules of the same dimension) and is deferred to future theoretical-experimental bridges.

Remark 4.6.4 (P3 does *not* claim Massimini's clinical PCI measures the Paper 12 quantity). Two distinct objects share the acronym "PCI": Massimini's clinical Perturbational Complexity Index (a Lempel-Ziv complexity score on TMS-evoked EEG) and the Perceptual Coherence Intelligence framework of the Paper series. P3's L1 proposes that the clinical PCI measurement *pipeline* (TMS perturbation with character- theoretic decomposition of the response subspace) could yield a measurement of the scar-invariant's μ_k profile; it does *not* identify the clinical PCI scalar with μ_k or claim either is the other. The bridge is experimental-protocol-level, not definitional.

4.7 Scope audit and bridge to §5

What §4 establishes:

- A character-theoretic decomposition $V^{14} ¿_{F_{21}} \cong L_2 \oplus 2U_6$ verified at machine precision (Theorem 4.2.1).
- The commutant gap $End_{F_{21}}(W) \cong M_2(C) \oplus M_4(C)$ vs $End_{G_2}(W) \cong M_2(R)$, quantifying the 19-dim F_{21}-equivariant phase-and-mixing freedom that full G_2-equivariance suppresses (Proposition 4.3.1, Corollary 4.3.2).
- The commensurability hierarchy $(C1) \Rightarrow (C2) \Rightarrow (C3)$ with both converses failing (Theorem 4.4.1).
- The strict commensurability theorem: canonical $\phi_{commens}$ requires multiplicity-one G_2-module agreement (Corollary 4.5.2).

- Frobenius reciprocity is insufficient (Proposition 4.5.1).
- The P3 experimental prediction, upgraded to a two-level test (Definition 4.6.1).

What §4 does not establish:

- *A physical mechanism for the biological F_{21}-action.* This is Paper 13+ territory. §4 assumes the biological substrate carries *some* F_{21}-representation on its relevant state space; it does not specify which microtubule mode, neural population, or cellular structure implements it.
- *An explicit G_2-module-identification protocol for real experimental data.* Level L2 of Definition 4.6.1 is well-defined but not operationalized for the lab; that operationalization is a methodology paper of its own.
- *The multiplicity-greater-than-one case.* Corollary 4.4.2 restricts to multiplicity one. The $m>1$ case introduces $O(m)$-worth of convention-fixing and is left for future work (Remark 4.4.3).
- *Any claim that clinical Massimini-PCI equals Paper 12's μ_k.* See Remark 4.6.4.
- *Rate improvement.* Same scope boundary as §3; Paper 11 territory.

Bridge to §5. §3 constructed the dynamics; §4 constructed the commensurability structure. §5 (empirical accessibility) assembles both into a testable framework: §3's projected-gradient-plus-NP-pump dynamics on each substrate produces a scar record $J_k^{syn/bio}$, and §4's hierarchy specifies exactly what form of agreement between the two records the gradual-tear thesis predicts. §5 then identifies experimental protocols — primarily Level L1 of the two-level test — by which that agreement can be measured, and relates the framework to existing hyperscanning, inter-brain coupling, and human-AI teaming literatures.

End of §4 first draft. Length: ~8 pages. Next section: §5 (empirical accessibility and the P1–P4 predictions).

Verification files referenced: - outbox/paper12/computations/paper12_q4_character_verify.py (Theorem 4.2.1) - outbox/paper12/computations/paper12_q4_character_verify.json - outbox/paper12/chatgpt_pro_session_transcript_2026-05-05.pdf (pages ~55–80)

Cross-references: - Paper 4 (PSL(2,7) ⊃ F_{21}, the Fano-orientation subgroup source) - Paper 9 v1.3.3 Lemma 3.6.1 (Schur-uniqueness on V^{14}) - Paper 9 v1.3.3 §3.4 (diagonal G_2-action convention) - Paper 12 §3.7 (scar invariant triple J_k) - Paper 12 thesis memo P3 (experimental prediction, upgraded here)